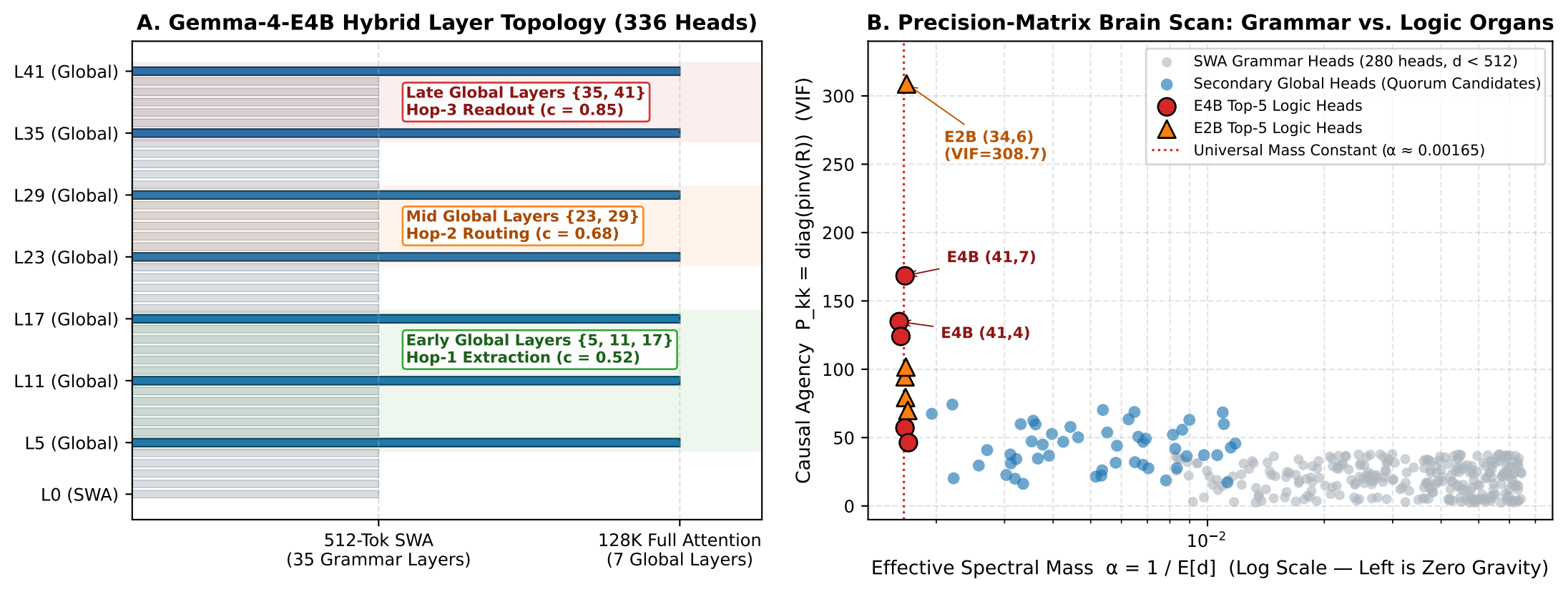
A. Gemma-4-E4B Hybrid Layer Topology (336 Heads)
L41 (Global)
L35 (Global)
L29 (Global)
L23 (Global)
L17 (Global)
L11 (Global)
L5 (Global)
L0 (SWA)
Late Global Layers {35, 41}
Hop-3 Readout (c = 0.85)
Mid Global Layers {23, 29}
Hop-2 Routing (c = 0.68)
Early Global Layers {5, 11, 17}
Hop-1 Extraction (c = 0.52)
512-Tok SWA
(35 Grammar Layers)
128K Full Attention
(7 Global Layers)
B. Precision-Matrix Brain Scan: Grammar vs. Logic Organs
SWA Grammar Heads (280 heads, d < 512)
Secondary Global Heads (Quorum Candidates)
E4B Top-5 Logic Heads
E2B Top-5 Logic Heads
Universal Mass Constant (α ≈ 0.00165)
E2B (34,6)
(VIF=308.7)
E4B (41,7)
E4B (41,4)
Causal Agency P_kk = diag(pinv(R)) (VIF)
0
50
100
150
200
250
300
10^{-2}
Effective Spectral Mass α = 1 / E[d] (Log Scale — Left is Zero Gravity)